Random Machine Learning Concepts

Two-Page Quick Report

This report covers Gradient Descent, Regularization, and the Bias–Variance Tradeoff.

1. Gradient Descent

Gradient Descent is an optimization algorithm used to minimize a loss function. Instead of directly finding the best parameters, it repeatedly updates model parameters in the direction that reduces the loss.

Basic idea: Calculate the gradient → determine the direction of greatest increase → move parameters in the opposite direction → repeat until the loss is sufficiently small or a stopping condition is reached.

Learning Rate

The learning rate controls the size of each parameter update. A very small learning rate can make training slow, while a very large learning rate can cause unstable updates or prevent convergence.

Applications

• Linear and logistic regression optimization

• Neural network training

• Many other differentiable machine learning models

2. Regularization

Regularization adds a penalty to the model's objective to discourage unnecessarily complex parameter values. Its main purpose is to reduce overfitting and improve generalization.

L1 Regularization

L1 regularization adds a penalty based on the absolute values of coefficients. It can push some coefficients exactly to zero, which can make it useful for feature selection.

L2 Regularization

L2 regularization adds a penalty based on squared coefficient values. It generally shrinks coefficients toward zero without typically making them exactly zero.

3. Bias–Variance Tradeoff

Bias and variance describe two different sources of prediction error. Understanding their relationship helps explain why a model that is too simple or too complex can perform poorly.

High Bias

A high-bias model is usually too simple to capture important patterns in the data. It may underfit both the training data and unseen data.

High Variance

A high-variance model is highly sensitive to the particular training dataset. It may fit training data extremely well while performing poorly on unseen examples, which is a common sign of overfitting.

Finding a Balance

The goal is not to minimize bias or variance independently. The goal is to find a model complexity that captures meaningful patterns while remaining robust to new data.

• More complex models often reduce bias but can increase variance.

• Simpler models often reduce variance but can increase bias.

• Cross-validation helps compare models and tune complexity.

• Regularization is one practical method for controlling model complexity.

4. Train, Validation, and Test Data

A dataset is commonly divided into training, validation, and test sets. The training set is used to learn parameters, the validation set helps select models or hyperparameters, and the test set provides a final estimate of performance on unseen data.

The test set should remain untouched during model development. Repeatedly checking the test set while making modeling decisions can lead to indirect overfitting to the test data.

Key Takeaway

Machine learning is not only about fitting a model. Optimization determines how parameters are learned, regularization controls complexity, and validation helps estimate whether the learned patterns will generalize beyond the training data.